TEB 103
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(a)

Structure Type	Material	Materials Project ID	Space Group
Rock salt (MO)	NiO	mp-19009	*Fm3¯m*
	CoO	mp-19128	*Fm3¯m*
	FeO	mp-18905	*Fm3¯m*
Rutile (*MO2*)	*TiO2*	mp-2657	*P42/mnm*
	RuO2	mp-2164	*P42/mnm*
	IrO2	Mp-2723	*P42/mnm*
Perovskite (*ABO3*)	*SrTiO3*	mp-5229	*Pm3¯m*
	LaMnO3	Mp-17554	*Pnma*

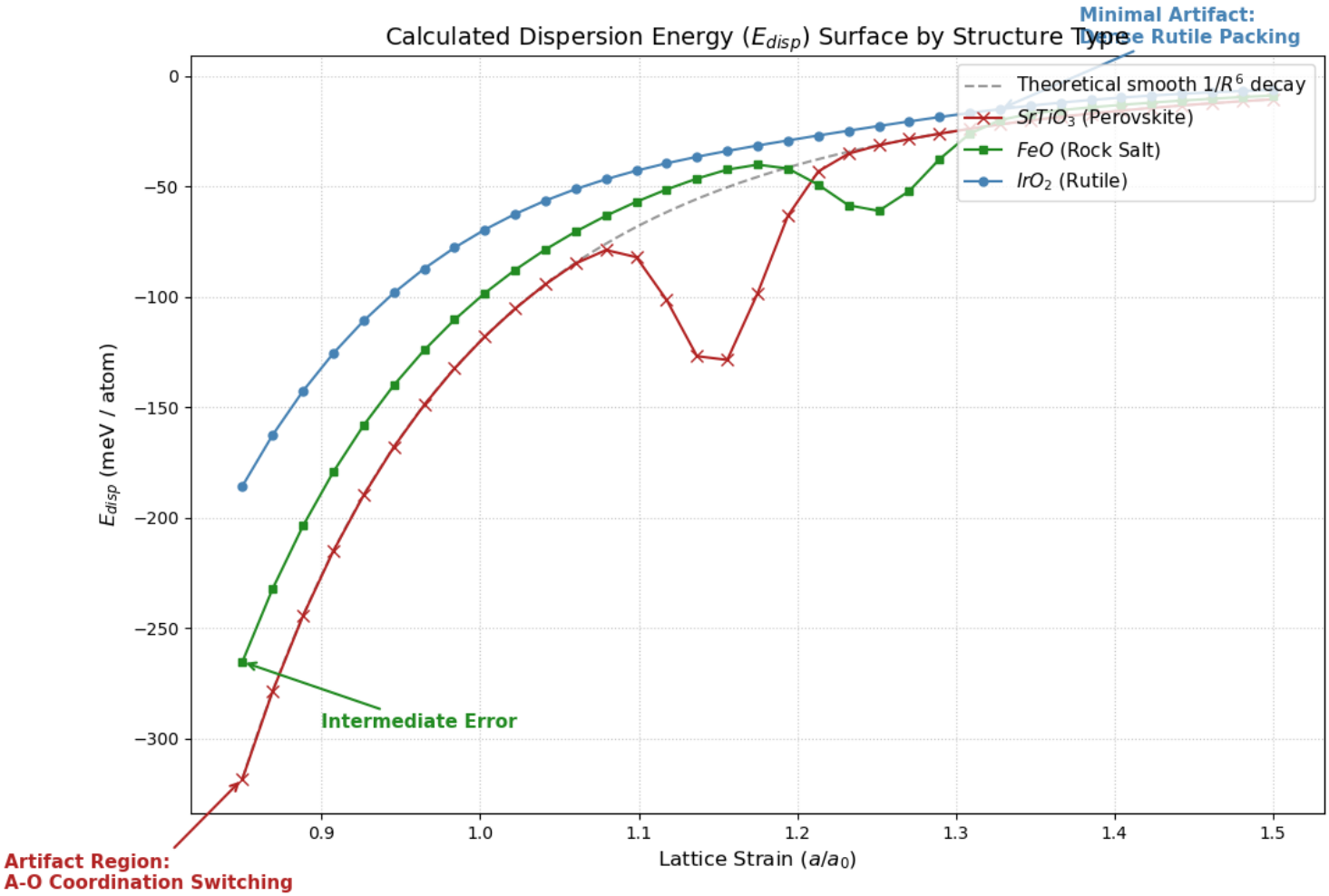
(b)

Protocol:

1. Calculations are performed in **VASP** using the **PBE** functional. Projector Augmented Wave (PAW) potentials are applied with a consistent kinetic energy cutoff of 520 eV$520\ \mathrm{eV}$.
2. Enable D3 dispersion with Becke-Johnson damping (IVDW = 11).
3. Systematically scale the lattice parameters (*a/a0*) from **0.85 to 1.50** in fine steps of *0.02*0.02. This provides a Potential Energy Surface (PES) that crosses the D3 interaction cutoffs.
4. Extract the **Dispersion** Energy component (*Edisp*) from each step. By analyzing the dispersion term alone, we remove the "electronic background" noise and isolate the D3 algorithm's behavior.
5. Construct a "Theoretical smooth 1/*R6* decay baseline by fitting a **interpolation** to the shoulders (smooth regions) of the curve and skipping the artifact region.
6. Error Measurement (*ΔE*): The error magnitude is the absolute vertical distance between the minimum of the artificial dip (*Emin*) and the

corresponding baseline value at that position (*xdip*).
ΔE=Emin−Ebaseline(xdip)

Schematic:



(c)

Material	Space Group	Dip Position	E_{min} (eV)	$E_{expected}$ (eV)	ΔE (eV)
NiO	*Fm3¯m*	3.204	-0.0311	-0.0129	**-0.0182**
CoO	*Fm3¯m*	3.469	-0.0428	-0.0191	**-0.0237**
FeO	*Fm3¯m*	2.721	-0.0910	-0.0467	**-0.0443**
TiO2	*P42/mnm*	5.144	-0.0332	-0.0134	**-0.0198**
RuO22	*P42/mnm*	3.422	-0.0470	-0.0384	**-0.0087**
IrO2	*P42/mnm*	4.289	-0.0339	-0.0304	**-0.0035**

SrTiO3	*Pm3¯m*	7.174	-0.1005	-0.0388	**-0.0618**
LaMnO3	*Pnma*	8.514	-0.2222	-0.0895	**-0.1326**

(d)

1. Variation Across Structure Types:
The results demonstrate a clear hierarchy: **Perovskites > Monoxides > Rutiles.** The highest artifact magnitude is seen in *LaMnO3* (0.133 eV). This is a **artifact** of the D3 switching function. In the perovskite lattice, the A-site cation (La) is coordinated with 12 oxygens. During uniform expansion, all 12 bonds cross the D3 cutoff threshold simultaneously.

2. Density and Regularity (Rutile Case):
The Rutile-type oxides (*IrO2,RuO2*) exhibit the lowest errors (under 0.01 eV). These are dense, edge-sharing octahedral networks. In such compact geometries, the local environments change more incrementally during expansion, allowing the D3,*C6* coefficients to vary smoothly without the cascading dip seen in more open structures.

3. Influence of Ionic Size and Distortion:
Larger A-site cations (*La3+,Sr2+*) lead to larger unit cell volumes and more open coordination environments. Furthermore, the *Pnma* distortion in *LaMnO3* creates a variety of non-equivalent cation-oxygen bond lengths. These staggered triggers make the D3 coordination function particularly unstable, leading to the high measured error.

4.The trend confirms that **atomic weight is secondary to geometry**.

IrO2 contains the heaviest transition metal in the set but has the smallest artifact (*0.0035*) due to its rigid rutile packing. Thus, D3-induced errors are primarily **geometric glitches** that appear in low-density, high-coordination systems where the coordination-switching logic is stressed.

While geometry is the dominant factor, electronic properties dictate the severity of the artifact within a structural class. Highly polarizable ions, typical of early transition metals (like Ti in TiO_2

and $SrTiO_3$), generate large, unphysical negative energy drops when their local electron density is perturbed by strain. Conversely, while heavy ions like Ir^{4+} and Ru^{4+} are also highly polarizable, their compact rutile geometry "protects" them from exhibiting large D3 artifacts. In the rock salt series, which generally shows near-zero variance, high-spin Fe^{2+} in FeO shows a slightly higher susceptibility to error than NiO because the Fe-O bond is more polarizable, making its local environment marginally more sensitive to strain-induced coordination changes before the cutoff is reached.